WORKED SOLUTION

STEP III 2006, Question 6

Pure • Polar Curves and Reflection

STEP III 2006, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q6

The key idea

The gradient formula is just the chain rule applied to $x = r\cos\theta$, $y = r\sin\theta$. For the mirror, put the pole at O with the initial line along L: the ray reflected at a point P then runs straight back along the radius, so its direction is given by θ alone. The law of reflection says the normal bisects the angle between the two rays, which pins the normal (and so the tangent) down in terms of θ. That turns the geometry into a separable differential equation for $r(\theta)$.

Solution

The gradient in polar coordinates

A curve given by $r = r(\theta)$ has Cartesian coordinates $x = r\cos\theta$, $y = r\sin\theta$, with r a function of θ. Differentiating with respect to θ by the product rule,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta.$$

By the chain rule, wherever $\mathrm{d}x/\mathrm{d}\theta \neq 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\tan\theta + r}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta} - r\tan\theta},$$

dividing top and bottom by $\cos\theta$ (allowed when $\cos\theta \neq 0$). This is the required gradient.

It is worth keeping the undivided version too: the vector

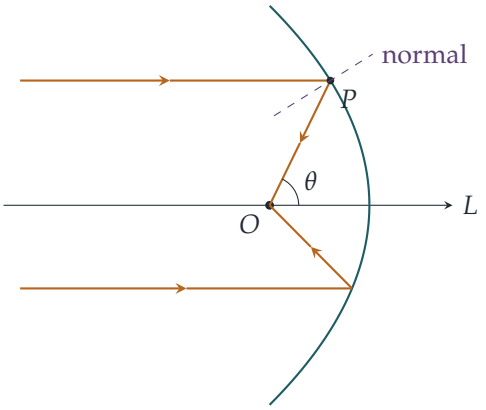
$$\mathbf{t} = \left(\tfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta,\ \tfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta\right)$$

points along the tangent at every point of the curve, including where $\tan\theta$ or the gradient is undefined.

The mirror

Take any plane Π containing L, and let P be a point where the mirror meets Π. The incident ray at P is parallel to L, so it lies in Π; the reflected ray passes through O, which is on L, so it lies in Π too. The law of reflection also puts the normal at P in the plane of the two rays, so the normal lies in Π, and the tangent to the curve of intersection at P is the line in Π perpendicular to it. So everything happens in the plane Π, and we can work in two dimensions.

In Π use polar coordinates with pole O and initial line along L, oriented so that the incident light travels in the direction $\theta = 0$ (left to right in the figure). Let P be (r, θ) with $-\pi < \theta < \pi$. (The point $\theta = \pi$ would be on L behind O, where a ray would pass straight through O without needing a mirror.)



The direction of the normal. At P take the two unit vectors pointing away from P along the rays:

$$\mathbf{a} = (-1, 0) \quad \text{(back along the incident ray)}, \qquad \mathbf{b} = (-\cos\theta, -\sin\theta) \quad \text{(towards } O\text{)}.$$

The normal makes equal angles with $\mathbf{a}$ and $\mathbf{b}$, so it lies along the bisector of the angle between them, which is the direction of $\mathbf{a} + \mathbf{b}$ (the diagonal of the rhombus formed by two unit vectors bisects their angle). Now

$$\mathbf{a} + \mathbf{b} = -(1 + \cos\theta,\ \sin\theta) = -2\cos\tfrac{\theta}{2}\left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\right),$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. Since $-\pi < \theta < \pi$, $\cos\frac{\theta}{2} > 0$, so this is non-zero and the normal at P is along

$$\mathbf{n} = \left(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}\right) :$$

the normal makes angle $\frac{\theta}{2}$ with L.

Ask yourself Why not just set the gradient formula equal to the gradient of the tangent?

We can: the tangent is perpendicular to $\mathbf{n}$, so its gradient is $-\cot\frac{\theta}{2}$, and equating this to the polar formula and clearing fractions leads to the same equation as below. But that route divides by $\tan\theta$, $\cot\frac{\theta}{2}$ and the denominator of the formula, so it breaks at $\theta = 0$ and $\theta = \pm\frac{\pi}{2}$. The tangent vector $\mathbf{t}$ avoids all of that.

The differential equation. The tangent is perpendicular to the normal, so $\mathbf{t}\cdot\mathbf{n} = 0$:

$$\frac{\mathrm{d}r}{\mathrm{d}\theta}\left(\cos\theta\cos\tfrac{\theta}{2} + \sin\theta\sin\tfrac{\theta}{2}\right) + r\left(\cos\theta\sin\tfrac{\theta}{2} - \sin\theta\cos\tfrac{\theta}{2}\right) = 0.$$

By the compound-angle formulae the brackets are $\cos\left(\theta - \frac{\theta}{2}\right) = \cos\frac{\theta}{2}$ and $\sin\left(\frac{\theta}{2} - \theta\right) = -\sin\frac{\theta}{2}$. So

$$\frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\tfrac{\theta}{2} = r\sin\tfrac{\theta}{2}, \qquad \text{that is,} \qquad \frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \tan\tfrac{\theta}{2},$$

dividing by $r\cos\frac{\theta}{2}$, which is positive because $P \neq O$ and $-\pi < \theta < \pi$.

Solving it. The equation is separable. Integrating with respect to θ, and noting $\int \tan\frac{\theta}{2}\,\mathrm{d}\theta = -2\ln\cos\frac{\theta}{2}$ (valid as $\cos\frac{\theta}{2} > 0$),

$$\ln r = -2\ln\cos\tfrac{\theta}{2} + C \quad\Longrightarrow\quad r\cos^2\tfrac{\theta}{2} = k,$$

where $k = \mathrm{e}^C > 0$ is constant along the curve (the curve is a single connected piece $r = r(\theta)$ on the interval $-\pi < \theta < \pi$). Using $2\cos^2\frac{\theta}{2} = 1 + \cos\theta$,

$$r(1 + \cos\theta) = 2k.$$

Converting to Cartesians. With $x = r\cos\theta$ and $r = \sqrt{x^2 + y^2}$ this reads $\sqrt{x^2 + y^2} = 2k - x$. Squaring (both sides are non-negative, since $2k - x = r \geq 0$),

$$x^2 + y^2 = 4k^2 - 4kx + x^2 \qquad\Longrightarrow\qquad y^2 = 4k(k - x).$$

This is a parabola with axis L, vertex at $(k, 0)$ and focus at the origin O, opening back towards the incoming light. Since Π was any plane containing L, the mirror meets every such plane in a parabola, as required. ■

Checks

- Run it backwards: on $y^2 = 4k(k - x)$ the normal is along $(4k, 2y)$; reflecting the direction $(1, 0)$ in this normal gives a ray through O. We checked this numerically at 10^4 random points with $k = 1.3$, and also checked that $r = 2k/(1 + \cos\theta)$ satisfies $r'/r = \tan\frac{\theta}{2}$ by finite differences.
- The gradient formula agreed with a finite-difference gradient of random curves $r(\theta)$ to about 10^{-8}.
- At $\theta = 0$ the normal lies along L and the tangent is perpendicular to L: that is the vertex $(k, 0)$, where a ray along L reflects straight back through O.

Common slips

- Using the external bisector $\mathbf{a} - \mathbf{b}$ for the normal. That direction makes supplementary angles with the two rays, not equal ones, and leads to $r'/r = -\cot\frac{\theta}{2}$, whose solution $r(1-\cos\theta) = 2k$ is a parabola facing the other way, focusing light that arrives from the opposite direction.
- Forgetting to explain why the problem is two-dimensional: the normal must lie in the plane containing L and P, or the curve of intersection need not have the tangent we used.
- Squaring $\sqrt{x^2+y^2} = 2k - x$ without noting both sides are non-negative.

Beyond STEP

The mirror is a paraboloid of revolution, and the property proved here is the focus–directrix property in disguise: $r(1+\cos\theta) = 2k$ says that the distance from O equals the distance from the line $x = 2k$. This is why satellite dishes, car headlamps and reflecting telescopes are paraboloids. More generally, conics all have the polar form $r(1+e\cos\theta) = \ell$, which is exactly what appears in Newton's solution of planetary orbits; Kleppner and Kolenkow, *An Introduction to Mechanics*, derives it from the inverse-square law.

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